Can a High School Personal Finance Course Reduce the Gender Gap in Being Unbanked?

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August 8, 2026

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Abstract

Among unpartnered young adults in the United States—whose banking status reflects their own financial decisions rather than a partner's—women are significantly more likely than men to be unbanked, even after accounting for income and education. I determine whether state-mandated high school personal finance courses reduce this gender gap, and if so, through what mechanisms. Using the FDIC National Survey of Unbanked and Underbanked Households (2009–2023) and a heterogeneity-robust difference-in-differences estimator, I estimate that mandates reduce the likelihood of being unbanked for both women and men, but by significantly more for women, narrowing the gender gap by 42.9 percent. The effects are concentrated among women with no education beyond high school and among Black and Hispanic women—groups that face the greatest barriers to financial inclusion. I examine two mechanisms, and the evidence supports both. First, the mandate increases women's objective financial knowledge, although not their self-assessed knowledge. Second, it improves economic outcomes by increasing college completion, self-employment, and the likelihood of attaining middle-income status. Because it reaches individuals before financial independence and does not rely on voluntary participation, mandatory high school financial education can reduce persistent gender disparities in financial inclusion that adult financial education programs have struggled to address.

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1 Introduction

Bank accounts are important gateways to participation in the formal financial system. They allow households to receive income, make payments, accumulate savings, establish credit histories, and access mainstream financial products. However, despite the widespread availability of banking services in the United States, approximately 4.2 percent of households, or about 5.6 million households, remained unbanked in 2023.[1] Among young adults without partners, the unbanked share rises to 10.3 percent. Moreover, within this subgroup, where banking decisions largely reflect individuals' own financial choices rather than those of a partner, a staggering 13.2 percent of women are unbanked, compared to 7.0 percent of men. Much of this gap reflects the fact that single women tend to be worse off financially than their male counterparts. However, even after accounting for differences in income, education, and other observable characteristics, a substantial gap remains, suggesting that factors beyond economic resources continue to limit women's access to the banking system. Understanding the sources of this persistent disparity is important for identifying policies that can effectively promote financial inclusion.

In this paper, I ask whether public policy can reduce the gender gap in financial inclusion, by studying the effects of mandatory high school personal finance education on the likelihood of being unbanked. Existing research describes who is unbanked and why (Bogan and Wolfolds, 2022; Hayashi, Routh and Toh, 2024), but much less is known about the policies that could close these gaps. One reason to expect financial education to help is that opening and maintaining a bank account requires understanding financial products, comparing account options, and navigating financial institutions, so improving financial knowledge before adulthood can raise banking participation and narrow the gender gap.

The analysis combines a descriptive decomposition with a causal research design. I first examine how much of the gender gap in being unbanked can be explained by differences between unpartnered women and men in income, education, household structure, and other

[1] Author's calculations using the 2023 FDIC National Survey of Unbanked and Underbanked Households.

observable characteristics. Using a Kitagawa–Oaxaca–Blinder decomposition of the gender gap in unbanked status with data from the Federal Deposit Insurance Corporation (FDIC) survey, I quantify the contribution of each characteristic. Because this decomposition is descriptive and cannot establish whether changing these characteristics would narrow the gap, I then turn to a specific policy: state mandates requiring students to complete a personal finance course before graduating from high school. Unlike voluntary financial education programs for adults, these mandates reach nearly all students before they become financially independent, avoiding concerns about selective participation. Variation in the timing of mandate adoption across states and graduating cohorts provides the basis for identifying the policy's effect. I combine information on mandate timing with data from the 2009 through 2023 waves of the FDIC National Survey of Unbanked and Underbanked Households and estimate the policy's effect using the extended two-way fixed-effects estimator of Wooldridge (2025), which accommodates heterogeneous treatment effects under staggered policy adoption.

I then examine the mechanisms through which the mandates affect banking outcomes. One possibility is that they improve financial literacy, making young adults better equipped to open and manage bank accounts. Another is that they improve economic circumstances by increasing educational attainment, employment, and income, thereby making bank account ownership more accessible and valuable. I examine the financial literacy channel using the National Financial Capability Study, which includes the Big Five financial literacy questions absent from the FDIC survey, and the economic channel by estimating the mandates' effects on educational attainment, employment, and income using the FDIC survey.

I find that mandatory high school personal finance education narrows the gender gap in being unbanked. Exposure lowers men's probability of being unbanked by 1.8 percentage points, or 25.3 percent of the rate among men, and women's by 4.4 percentage points, or 33.5 percent of the rate among women, closing 42.9 percent of the gender gap. The effects are concentrated among women with only a high school education and among Black and

Hispanic women, the groups furthest from the formal financial system to begin with. The evidence supports both mechanisms. Relative to men, exposure increases women's financial knowledge scores by 0.13 points, 5.9 percent of the mean, with the gains concentrated in the two investment questions. In contrast, it has no effect on women's self-assessed financial knowledge, suggesting that what changes is what they know rather than what they believe they know. Women's economic circumstances also improve. Relative to men, exposure makes women 2.9 percentage points more likely to complete college, 1.3 percentage points more likely to be self-employed without incorporating, and 3.9 percentage points more likely to have income between $50,000 and $75,000.

These findings suggest that mandatory high school personal finance education can be an effective policy tool for expanding financial inclusion among young adults by reducing barriers to banking participation. The larger gains among women indicate that broad-based financial education can also narrow gender disparities in financial inclusion. At the same time, the weaker effects for men suggest that a uniform curriculum may not benefit all students equally, highlighting the importance of accounting for heterogeneous responses when designing financial education policies.

2 Related Literature

The first body of research examines the causes and consequences of being unbanked. This literature has established that households without a bank account are disproportionately low-income, less educated, and members of racial and ethnic minority groups (Boel and Zimmerman, 2022; Calem, Henderson and Wang, 2026). Less is known about how these patterns translate into gender disparities in financial inclusion. Bogan and Wolfolds (2022) show that gender differences in being unbanked vary substantially across racial and ethnic groups: Black women are more likely to be unbanked than Black men and than any other race-gender group, Hispanic women are less likely to be unbanked than Hispanic men, and

White women and men have similar rates. Because existing measures of banking status are collected at the household level, however, these comparisons may reflect both individual financial decisions and differences in who reports household financial information. Focusing on unpartnered adults, for whom household banking status corresponds more closely to individual banking participation, I document in Section 5 that women are substantially more likely than men to be unbanked.

A growing literature examines the factors underlying these disparities and the potential for policy to reduce them. Calem, Henderson and Wang (2026) show that the decline in the overall unbanked rate has coincided with rising incomes, but that remaining unbanked households are increasingly concentrated among single individuals. They also find that states with higher levels of financial literacy have lower unbanked rates, suggesting that financial capability may be an important determinant of banking participation. However, evidence on whether policies that improve financial knowledge can expand financial inclusion remains limited. Policies that focus solely on increasing access to bank accounts have generally produced limited long-run effects (Dupas et al., 2018; Collins, Larrimore and Urban, 2026), suggesting that demand-side barriers may play an important role. Such barriers include limited financial knowledge and distrust of financial institutions (Hayashi, Routh and Toh, 2024). They are not all of one kind. Some are preference-based, in that the household does not want an account. Others are informational, in that the household wants one but does not know which products carry low or no minimum balances, or how to open and maintain one. Policies that expand access address neither, and an education mandate can address only the second. This paper contributes to this literature by examining whether a policy designed to improve financial knowledge, mandatory high school personal finance education, increases banking participation and whether its effects differ across demographic groups.

The second body of research examines financial literacy and, in particular, gender gaps in financial literacy. Across countries, women score lower than men on standard measures of financial knowledge, are more likely to respond "don't know" rather than guess, and report lower confidence in their financial knowledge even conditional on measured knowledge (Lusardi and Mitchell, 2011; Bucher-Koenen et al., 2017; Aristei and Gallo, 2022). This gap cannot be fully explained by differences in observable characteristics. Decomposition analyses show that differences in income, education, and other demographic characteristics account for only a small share of the gender gap, with much of the remaining difference reflecting variation in the relationship between these characteristics and financial knowledge (Fonseca et al., 2012; Aristei and Gallo, 2022). Consistent with this interpretation, Fonseca et al. (2012) show that financial decision-making within couples does not follow a simple specialization model in which one spouse manages household finances, but instead depends on the relative education of partners. Aristei and Gallo (2022) further suggest that behavioral traits and social norms regarding gender roles in financial decisions contribute to the remaining gap. Importantly, gender differences in financial literacy persist among groups for whom individual financial knowledge is particularly consequential, including single women and widows (Bucher-Koenen et al., 2017). These findings make financial literacy a particularly relevant demand-side barrier to study: it exhibits substantial gender disparities, is not merely a consequence of differences in income and education, and is potentially amenable to policy interventions through education. Whether improvements in financial literacy translate into greater financial inclusion, however, remains an open empirical question.

The third body of research examines the effects of state-mandated high school financial education. Early evaluations of these mandates often found limited effects on subsequent financial knowledge and behavior (Cole, Paulson and Shastry, 2016; Fernandes, Lynch and Netemeyer, 2014; Willis, 2011). More recent evidence, however, has produced more positive findings by leveraging the staggered adoption of mandates across states and cohorts. This literature shows that exposure to high school financial education increases financial knowledge and improves a range of downstream financial outcomes, including credit management, borrowing decisions, debt outcomes, and reliance on alternative financial services (Walstad, Rebeck and MacDonald, 2010; Kaiser and Menkhoff, 2020; Urban et al., 2020; Stoddard and Urban, 2020; Brown et al., 2016; Harvey, 2019). Evidence from randomized evaluations outside the United States similarly finds that school-based financial education improves both financial knowledge and financial behaviors (Bruhn et al., 2016, 2025; Frisancho, 2023). Together, these studies demonstrate that financial education mandates can influence financial capability and a broad range of financial outcomes. However, less is known about whether such policies increase participation in the formal banking system and whether their effects differ across demographic groups.

A small but growing literature examines the relationship between financial education and banking participation directly. Harvey (2019) finds that financial education reduces young adults' reliance on alternative financial services, partly through improvements in financial literacy and financial planning. More recently, Routh and Urban (2025) use the FDIC National Survey of Unbanked and Underbanked Households to show that state financial education mandates reduce the likelihood of being unbanked, particularly in states with stronger payday lending restrictions. Their findings highlight the potential role of financial education in expanding financial inclusion, but their analysis focuses on the interaction between financial education and payday lending regulation rather than on the direct effects of education mandates on banking participation. This paper builds on this literature by examining whether state-mandated financial education increases banking participation and whether these effects contribute to reducing gender disparities in financial inclusion.

My study addresses these gaps in three ways. First, I examine whether high school financial education reduces gender disparities in banking participation, rather than simply lowering the overall unbanked rate. To better capture individuals' own financial decisions, I focus on unpartnered young adults, for whom household banking status more closely corresponds to individual banking participation.[2] Second, I examine which groups benefit most from the policy and show that the largest gains are concentrated among women with no education

[2]Because unbanked status is measured at the household level, married or cohabiting respondents share the same banking status as their partners. As a result, gender differences among partnered individuals may not reflect differences in individual financial decisions. I discuss this issue further in Section 4.

beyond high school and among Black and Hispanic women, groups with the lowest baseline rates of banking participation. Third, I explore why financial education has larger effects on banking participation among women than men. Using the National Financial Capability Study, I show that mandate exposure increases women's objective financial knowledge but does not improve their self-assessed knowledge, consistent with the distinction between financial knowledge and confidence emphasized by Aristei and Gallo (2022). Using the FDIC survey, I further show that mandate exposure is associated with improvements in women's socioeconomic outcomes through higher college completion, greater self-employment, and movement into the middle of the income distribution.[3] Together, these contributions extend the literature from asking whether financial education affects financial outcomes to understanding whether it can reduce demographic disparities in financial inclusion, which groups benefit most, and the mechanisms underlying these effects.

3 Institutional Background

Over the past several decades, states have increasingly adopted legislation requiring high school students to complete personal finance coursework before graduation. By 2030, 42 states will have implemented such requirements. These mandates were designed to improve students' financial knowledge and prepare them to manage financial decisions as they transition into adulthood. Personal finance courses are typically offered in grades 10 through 12, with most students taking them during their junior or senior year, and cover topics such as budgeting, saving, *banking*, credit, debt management, insurance, and investing. The banking component is directly relevant to the outcome studied here: it teaches students how to compare account fees, select among financial products, open accounts, and avoid costly mistakes such as overdrafts. By providing this information before students begin managing their own finances, personal finance education may influence whether young adults enter adulthood

[3] This finding is consistent with Struckell et al. (2022), who find a stronger positive relationship between financial literacy and self-employment among women than among men.

with a connection to the formal banking system.

These state mandates provide a useful setting for studying whether financial education affects banking participation because exposure is determined by state graduation requirements rather than individual decisions to enroll in a course. All students in an affected graduating cohort must complete the required instruction to graduate, including students who might not otherwise receive financial education at school or at home. This feature distinguishes mandatory financial education from voluntary programs, where participation may reflect unobserved differences in financial interest or ability.

A second feature of these mandates is their timing. Although some students may already have experience using a bank account before completing high school, the transition out of high school is when individuals increasingly begin managing finances independently, including earning income, paying bills, and making their own financial decisions. Personal finance courses are typically completed shortly before this transition, allowing the instruction to shape financial behaviors as young adults begin managing their finances independently.

States adopted these requirements at different times, generating variation in exposure across states and graduating cohorts. Illinois is the earliest adopter, with the requirement first applying to the 1970 graduating cohort. 3 additional states first implemented requirements for cohorts graduating in the 1990s, followed by 11 states in the 2000s and 16 states in the 2010s, with remaining states adopting in the 2020s and beyond. A summary of the timing of adoption and the first graduating cohort exposed to each requirement is provided in Table A1.

States differ in how they implement personal finance graduation requirements. Some require students to complete a stand-alone personal finance course, while others integrate personal finance instruction into existing courses such as economics, consumer education, or civics. As shown in Table A1, stand-alone requirements are uncommon over the period I study. Among the 31 states contributing treated cohorts, only 5 require a stand-alone course by the 2019 cohort, while 26 provide instruction through an embedded curriculum. Most stand-alone requirements were adopted after the cohorts observed in my sample. I therefore define treatment as whether a state requires personal finance instruction for graduation, regardless of whether the instruction is delivered through a stand-alone course or an embedded curriculum. I measure exposure using the first graduating cohort required to complete the mandate rather than the year of legislative adoption, because implementation often begins several years after legislation is enacted.

4 Data

I use two data sources. The primary dataset is the FDIC National Survey of Unbanked and Underbanked Households, which I use to estimate the effects of state personal finance mandates, examine treatment heterogeneity, and test the economic mechanism. I supplement this analysis with the National Financial Capability Study (NFCS), which contains detailed measures of financial knowledge, including the Big Five financial literacy quiz. Although the NFCS records respondents' ages only in broad categories, making it less suitable for the main identification strategy, it is what allows me to test whether the mandates improve financial knowledge.

4.1 FDIC National Survey of Unbanked and Underbanked Households

The FDIC survey is a nationally representative supplement to the Current Population Survey (CPS), fielded every two years since 2009, and I use all eight waves available at the time of writing, 2009 through 2023. The survey is the leading source of information on banking participation in the United States, and it records household banking status together with rich demographic and socioeconomic characteristics. Because it records each respondent's state of residence and age, it can be linked to the timing of state personal finance mandates, under assumptions about where respondents attended high school that I set out below.

Sample selection criteria. The supplement is answered by one designated respondent per household, who reports banking status on the household's behalf, so each observation is both a respondent and a household. Across the 2009 through 2023 waves, 296,005 households completed the supplement, out of 570,943 CPS households in the months it was fielded. I reduce these to the 24,955 respondents used in estimation by imposing five criteria in the following order. First, I drop 28,581 respondents who did not complete high school, because treatment is defined by exposure to a graduation requirement and those who left school before graduating were never subject to one. Second, I drop 178,613 respondents outside the 1991 through 2019 high school graduating cohorts, because cohorts graduating earlier experienced different underlying trends and make less comparable controls, and cohorts graduating later are not yet observed as independent adults. Third, I drop 30,320 respondents observed outside ages 22 to 36, of whom 3,309 are younger and 27,011 older. Fourth, I drop 923 respondents who did not report family income, so that a single sample underlies the difference-in-differences estimates, the decomposition, and the analysis of income outcomes. Fifth, I drop 32,613 respondents who live with a spouse or partner, because their banking status is shared with a partner, so a gender comparison among them reflects who was surveyed rather than their own banking decisions.

Because exposure is assigned by state and graduation cohort, it is variation across these cohorts and states, and not variation in age, that identifies the effect of the mandate. I set the two age bounds for separate reasons. I set the lower bound at 22 because of the household-level measurement noted above: respondents younger than 22 are often still enrolled in school and living with their parents, so their recorded status describes their parents' banking rather than their own. I set the upper bound at 36 to hold the observation window comparable across cohorts, since the cohort restriction on its own would admit the earliest cohorts at much older ages. A respondent who graduated in 1991 is 50 by the 2023 wave, and the cap keeps every cohort observed over the same span of early adulthood.

Restricting the sample to unpartnered individuals minimizes the influence of a partner's financial decisions on household banking status. I define unpartnered individuals as those living without a spouse or cohabiting partner, which counts respondents who are legally married but whose spouse is absent from the household as unpartnered. I impose this restriction because among partnered respondents a gender comparison reflects which partner answered the survey rather than a difference in their own banking decisions. Banking status is recorded for the household rather than the individual, so a partnered woman and her partner are assigned the same status, and surveying either one returns the same household rate.

Treatment assignment. Treatment is defined as exposure to a state-mandated high school personal finance course. I take the timing of each state's requirement, the first graduating cohort required to complete personal finance instruction, from Urban et al. (2020) and Urban's compilation of state financial education requirements, last updated in 2024.[4] The survey records the state a respondent lives in, not the state where they attended high school, so I assume the two coincide and that respondents graduated in the year they turned 18. Interstate migration between high school and the survey therefore assigns some respondents the wrong state's requirement, and the estimates are intent-to-treat with respect to the assigned requirement rather than to coursework actually completed. I consider a respondent to have been treated if their state implemented a mandate no later than the year of their high school graduation. I use this any-exposure rule, rather than requiring that the mandate be in place before the respondent entered high school, because most students take the required course toward the end of high school, as described in Section 3.

Outcome and covariates. The primary outcome is whether the respondent lives in an unbanked household, which the FDIC defines as a household in which no member has a checking or savings account at a bank or credit union.

The FDIC supplement records respondents' race and ethnicity, family income, education,

[4] Available at `https://papers.carlyurban.com/Policies_ForPosting_2024.xlsx`.

marital status, age, labor force status, and whether they were born in the United States. I recover respondents' gender and cohabitation status by linking the supplement to the underlying CPS files. I combine marital status, cohabitation status, and the presence of children younger than 18 to construct a measure of household structure. Educational attainment is classified into four categories: less than a high school diploma, high school graduate, some college, and college graduate. Family income is reported in five categories ranging from less than $15,000 to $75,000 or more. These are the survey's own categories, recorded in nominal dollars and not adjusted for inflation, so income enters the analysis as a set of category indicators rather than as a real amount. Family income includes the income of all household members related to the reference person by birth, marriage, or adoption. Therefore, income from an unmarried partner or an absent spouse is excluded. Because relatively few respondents identify as Asian, I combine them with other racial and ethnic groups into an "Other" category, leaving indicators for White, Black, Hispanic, and Other.

I compare respondents in ever-treated and never-treated states in appendix Table A2. Respondents in ever-treated states are more likely to be unbanked and have lower incomes and less education. These differences motivate the empirical strategy, which compares cohorts exposed and unexposed to the mandate within the same state rather than comparing states with one another.

4.2 National Financial Capability Study

The NFCS is conducted every three years by the Financial Industry Regulatory Authority (FINRA) Investor Education Foundation, and I use all six waves, fielded from 2009 to 2024. It records measures of financial literacy, financial attitudes, and exposure to financial education, none of which the FDIC survey collects.

The NFCS differs from the FDIC survey in two ways that bear on what I can ask of it. It is an online, non-probability sample rather than a nationally representative household survey, and it records age in broad bands rather than in single years, which is what assigning respondents to graduation cohorts requires. For both reasons I use the NFCS to examine the financial knowledge mechanism, and I leave the effect of the mandate on banking to the FDIC survey.

Exposure to financial education is recorded by a question asking whether a school, college, or workplace the respondent attended offered financial education, and whether the respondent took part. Respondents who did are then asked whether they received it in high school, in college, or from an employer. These items were first fielded in 2012, so exposure is measured in five of the six waves, while the financial knowledge questions are available in all six.

The primary outcome is financial knowledge, measured using the standard Big Five financial literacy questions developed by Lusardi and Mitchell (2011, 2014). Although a sixth question was introduced beginning in 2015, I use the original Big Five questions to maintain consistency across survey waves. I also examine self-assessed financial knowledge because studies have documented gender differences in financial confidence, even conditional on measured financial literacy (Bucher-Koenen et al., 2017; Aristei and Gallo, 2022).

To make the NFCS analysis comparable to the FDIC survey analysis, I restrict the sample to unpartnered high school graduates aged 18 to 44, which retains the youngest three of the six age bands the survey records (18–24, 25–34, and 35–44). Graduation cohorts can therefore be approximated only coarsely, and Section 6 sets out what that costs the design. The age floor of 18 raises no measurement problem here, because the NFCS records financial knowledge for the individual respondent rather than banking status for the household, so a respondent's answers are their own. The covariates are more limited than in the FDIC survey, with race observed only as White versus non-White respondents. The estimates therefore control for gender and for non-White status, alongside state, graduation-cohort, and survey-year fixed effects.

I report the same comparison for the NFCS sample in appendix Table A3. Respondents in ever-treated states have somewhat lower measured financial knowledge and less education, although the two groups report similar rates of financial education. The NFCS therefore supplies what the FDIC survey cannot, a direct measure of what students learned, at the cost of a coarser design.

5 Descriptive Patterns in the Gender Gap

The gender gap in being unbanked is concentrated among the groups least likely to hold an account at all, and locating where it is widest motivates the subgroup estimates that follow. In Table 1, I report the unbanked rate and the female−male gap across household structure, race and ethnicity, family income, education, and age. The gap is not uniform. It is widest among respondents with no education beyond high school, among Black and Hispanic respondents, and among those in the lowest income categories, and it is also large among never-married respondents with children. It does not vary with age, however: the gap is 5.87 percentage points at ages 22 to 26, 6.95 percentage points at ages 27 to 31, and 5.58 percentage points at ages 32 to 36, which weighs against reading it as a life-cycle phenomenon. In columns 5 through 7 of Table 6, I later estimate the effect of the mandate separately for these subgroups, and the estimated gender differentials are largest for education and for race and ethnicity.

I next ask whether women are more often found in the groups where the gap is widest, comparing the characteristics of women and men in the analysis sample in Table 2. As shown in the first row of Table 2, 13.24 percent of women lack a bank account, compared with 7.04 percent of men, so the rate among women is about 88 percent higher than among men. The two groups also differ along several dimensions. For example, women have lower family income and are substantially more likely to be single parents, whereas men in the sample are predominantly never-married without children. Women are also more likely to be out of the labor force and are more likely to be Black, which partly reflects the restriction to unpartnered respondents: 74.22 percent of Black women in this age and education window are unpartnered, compared with 40.16 percent of White women. At the same time, women are slightly more likely to have attended college, which would predict greater banking participation rather than a higher unbanked rate. With that one exception, these are the same groups in which the gap in Table 1 is widest: women are over-represented among never-married respondents with children, among Black respondents, and in the lowest income categories, while in education they are the better-educated group, so education works against the gap rather than toward it. These gender differences suggest that the raw gap partly reflects differences in economic circumstances and household structure, a possibility I quantify with the decomposition in Section 7.

Everything so far rests on characteristics the survey records. I turn now to what unbanked respondents say themselves, which bears on whether the gap reflects constraints women face or choices they make. In Table 3, I report the barriers they cite. In the first row, insufficient funds to meet minimum balance requirements is the most frequently cited barrier for both groups, and it is cited by 52.20 percent of unbanked women compared with 45.26 percent of unbanked men, a difference of 6.93 percentage points. The second row reverses the sign: 39.64 percent of men cite a lack of trust in banks, against 32.72 percent of women, a difference of 6.92 percentage points. These are the only two rows in which the gender difference is statistically significant. Women and men cite the remaining barriers at similar rates: fees (32.88 versus 37.01 percent), privacy (29.76 versus 31.93), identification (14.59 versus 17.45), product availability (14.93 versus 15.91), and inconvenient hours or locations (11.21 versus 14.44). The barrier women cite more often, not having enough money to meet a minimum balance, describes what they can afford, while the barrier men cite more often, distrust of banks, describes whether they want an account at all. Read that way, the gender gap in unbanked status is associated more with what women can afford than with whether they want to participate, which is also how the decomposition in Section 7 reads, since family income is the largest single contributor to the explained gap.

A natural question is whether unbanked women want a bank account less than unbanked men do. They do not. Among the 1,883 unbanked respondents observed in the 2009 through 2017 waves, which are the waves that carry the future-intentions questions, 45.92 percent of women report wanting to open an account, compared with 37.59 percent of men, while 42.57 percent of men report having never held an account and having no plans to open one, compared with 29.32 percent of women. Willingness to participate in the formal banking system is therefore not the binding constraint for unbanked women. What this rules out is the preference channel: if unbanked women simply did not want accounts, a course that supplies information could not move their banking status. Money is one constraint that does bind, since insufficient funds to meet minimum balance requirements is the barrier they report most often, and a high school course cannot relax it. What such a course can relax is information: which accounts carry low or no minimum balances, and how to open and maintain one. I test that channel in Section 7.4. These are stated intentions, however, and I cannot rule out that young low-income men rationalize an outcome as a choice while women more readily report wanting something they lack. I therefore read this pattern as the ex ante reason to expect the mandate to reduce unbanked rates more among women than among men, which is what I estimate in Section 7, rather than as evidence on the causal channel itself.

These descriptive patterns share a common thread. The gender gap in being unbanked is concentrated among women with fewer economic resources and more demanding household circumstances, and these women more often report wanting a bank account than report no interest in one. Two questions follow, which I address in the remainder of the study: how much of the gap these differences in circumstances account for, and whether a policy that builds financial literacy can narrow it. The next section describes the two analyses I use to answer them.

6 Empirical Strategy

My empirical analysis has two parts, and each answers a different question. The first part is descriptive. I ask how much of the female–male gap in being unbanked comes from the ways unpartnered women and men differ in income, education, household structure, and other characteristics. For this I estimate a linear probability model of unbanked status and use its coefficients in a Kitagawa–Oaxaca–Blinder decomposition, which splits the gap into the part that each characteristic accounts for. The second part is causal. I ask whether a state requirement to take personal finance in high school lowers the unbanked rate, and whether it lowers it more for women than for men and so narrows the gender gap in being unbanked. For this I use a staggered difference-in-differences design, estimated with the extended two-way fixed-effects estimator of Wooldridge (2025). I then estimate the same design on subgroups to ask which women the mandates help most. Finally, I test two of the mechanisms through which the mandates could narrow the gap: financial literacy, which the course is designed to raise, and the economic circumstances that make a bank account worth holding.

6.1 The Decomposition

The decomposition requires an estimate of how each characteristic is related to being unbanked, so I begin by fitting a linear probability model of unbanked status to the pooled sample of women and men, with income, education, labor-force status, household structure, race and ethnicity, whether the respondent was born in the United States, and age as covariates.[5] The explained component asks what would happen if women held men's characteristics while the relationship between characteristics and banking status stayed fixed. To

[5]Because the outcome is binary, a nonlinear decomposition such as Fairlie (2005) is also a natural choice. Fairlie's method is to estimate a logit model and decompose the difference in predicted probabilities rather than mean outcomes. In practice, however, nonlinear decompositions are more sensitive to sparse combinations of covariates because they rely on estimated predicted probabilities and repeated resampling to construct the decomposition. In my application, the extensive set of categorical covariates produce many thin cells, making the decomposition and the state-clustered bootstrap unstable across replications. Therefore, I use the standard Kitagawa–Oaxaca–Blinder decomposition based on a linear probability model, at the cost that fitted values are not constrained to lie between zero and one.

answer that, I value every characteristic at one coefficient common to women and men, which I take from the pooled model, following the pooled-model variant of Fortin, Lemieux and Firpo (2011). The pooled model includes an indicator for gender. Without it, the pooled coefficients would absorb part of the female–male differential itself, moving that part of the gap out of the unexplained component and into the explained one.

The decomposition (Kitagawa, 1955; Oaxaca, 1973; Blinder, 1973) then separates the gender gap in unbanked status into an explained component, arising from differences in observed characteristics, and an unexplained component, which is what remains when women and men with the same characteristics still differ in how likely they are to be unbanked. Each characteristic contributes the difference in its mean between women and men, multiplied by that characteristic's coefficient, so a characteristic accounts for more of the gap when women and men differ on it and when it is more strongly related to being unbanked. The explained component is the sum of those contributions, and it measures how much of the gap would close if women held the same characteristics as men. The unexplained component is the remainder, and it reflects differences between women and men in how strongly a given characteristic is related to being unbanked, together with characteristics the FDIC survey data do not measure.

These coefficients are estimated effects of the characteristics on the probability of being unbanked, although they are not causal ones. The estimated coefficient on family income, for example, might in part reflect financial education a respondent received earlier, or unobserved state, cohort, and individual factors that move income and banking status together, so I cannot conclude that raising women's incomes would close the share of the gap that the decomposition assigns to income. I do not use the decomposition for that purpose. I use it to establish which differences between women and men account for the observed gap, before turning to a policy that can be changed. What it finds also determines what I test later. Income, education, and employment are each strongly related to being unbanked, and differences in income account for more of the gap than differences in anything else I observe.

They are also the characteristics a high school course could plausibly move, which is why I take them up as the economic mechanisms in Section 7.4.

6.2 The Difference-in-Differences Design

The objective of this part of the analysis is to estimate the effect of the mandate on the probability of being unbanked, separately for women and for men, together with the difference between those two effects, which is the change in the gender gap. A simple comparison of unbanked rates between exposed and unexposed respondents would not recover these effects, because the states that adopted early differ from those that adopted late in ways that also affect banking status.

A difference-in-differences design is necessary because mandate exposure, although exogenous to the individual, is not exogenous to the state and cohort environment. Exposure is determined by the state where a respondent attended high school and the year of graduation, neither of which the respondent chooses. It is nonetheless correlated with unobserved determinants of banking status that vary across states and over time. Access to mainstream banking, for example, has changed unevenly across states through branch and ATM density, the availability of low-cost accounts, and the diffusion of mobile deposit.[6] Such unobservables are not recorded in the survey data, and they can vary with the timing of adoption.

I estimate the effect of the mandate using the extended two-way fixed-effects (ETWFE) estimator of Wooldridge (2025). Because states adopted in different years and the effect of the mandate can differ across adoption cohorts, the estimator must accommodate heterogeneous effects. ETWFE does so by estimating a separate treatment effect for every combination of adoption cohort and graduation cohort, drawing comparisons only against cohorts that are not yet treated or never treated, and averaging those effects into a single summary. A conventional two-way fixed-effects (TWFE) regression carrying a single treatment indicator

[6] Other examples include the assistance that caseworkers administering TANF and Medicaid provide to recipients in establishing direct deposit of benefits, and state and municipal account-opening initiatives such as the BankOn coalitions supported by the Cities for Financial Empowerment Fund, which have expanded at different rates across states.

instead uses already-treated cohorts as controls for later-treated ones, and such forbidden comparisons can receive negative weight in the estimated average effect (Goodman-Bacon, 2021).[7] A Goodman-Bacon decomposition indicates that forbidden comparisons carry only a small share of the weight here, which is why conventional TWFE suffices for the NFCS analysis, where the smaller sample and coarser cohort measure make the staggered-adoption estimators less feasible.

Let $s(i)$ denote respondent i's state and $c(i)$ their high school graduating cohort, defined as the year they turned 18. Section 3 describes the rollout and Table A1 lists each state's first exposed cohort. State s first requires the mandate for cohort g_s, with $g_s = \infty$ for never-treated states, so respondent i is exposed if $c(i) \geq g_{s(i)}$. Eleven states adopted mandates that first apply to cohorts graduating in 2020 or later, after the last cohort I observe, so respondents in those states are never exposed within the sample and serve as not-yet-treated controls. Let $D_i^{gt} = \mathbf{1}\{\, g_{s(i)} = g,\ c(i) = t,\ t \geq g \,\}$ indicate that respondent i belongs to treated cell (g, t), so that each treated respondent has $D_i^{gt} = 1$ for one cell only and untreated respondents have $D_i^{gt} = 0$ for every cell. I estimate

$$Y_i \;=\; \alpha_{g_{s(i)}} + \lambda_{c(i)} + X_i'\beta + \gamma\,\text{female}_i + \sum_g \sum_{t \geq g} \Big(\tau_{gt} + \delta_{gt}\,\text{female}_i\Big) D_i^{gt} + \varepsilon_i \tag{1}$$

by OLS, with standard errors clustered at the state level. Here Y_i is an indicator for being unbanked, α_g are adoption-cohort fixed effects, λ_t are graduation-cohort fixed effects, female_i is an indicator equal to one if respondent i is female, and X_i contains indicators for Black, Hispanic, and other race or ethnicity, with White the omitted category, together with an indicator for having been born outside the United States. These are the characteristics fixed before high school, and I return below to why the richer covariate list used in the decomposition does not belong here. The design applies to the FDIC survey even though it

[7] The Callaway–Sant'Anna and Sun–Abraham estimators also avoid them, and I report both alongside ETWFE in the event-study figure. I prefer ETWFE for the main estimates because it recovers all cohort-specific effects within a single pooled regression, so that the difference between women's and men's effects and the standard error of that difference come from one clustered covariance matrix.

is a repeated cross-section rather than a panel, because exposure varies at the level of the state and the graduating cohort rather than within individuals over time. To avoid sparse treatment cells, graduation cohorts are grouped into three-year bins.

Within a treated cell (g, t), τ_{gt} is the effect of the mandate on men, $\tau_{gt} + \delta_{gt}$ is the effect on women, and δ_{gt} is the difference between them, which is the change in the gender gap. Because there is one such set of parameters for every treated cell, I summarize them as averages over the treated. The average treatment effect on the treated is a count-share-weighted average of the cell-specific effects, each cell weighted by its share of treated respondents, and the female and male averages use the corresponding within-gender shares. These averages are what I report in Table 6: the Mandate exposure row is the average of τ_{gt}, the effect for men; the Mandate exposure $\times$ Female row is the average of δ_{gt}, the change in the gap; and the effect for women is the sum of the two. In column 1, I restrict δ_{gt} to zero and drop the covariates, which yields one average effect for the sample as a whole.

The two sets of fixed effects remove differences in levels. The adoption-cohort effects α_g absorb whatever is fixed over time and shared by the states that first required the course in the same year, and the graduation-cohort effects λ_t absorb whatever changed for all cohorts alike, including the national diffusion of mobile banking. Differences between two states that adopted in the same year are not absorbed by α_g, but they are differenced out, because a state's own level enters its cohorts both before and after the requirement binds and the estimator compares changes rather than levels. What identifies the effect of the mandate is the timing of adoption, which I plot in Figure 1: the staggered arrival of the requirement across graduating cohorts, and the share of the sample exposed in each cohort bin. What the fixed effects cannot remove is a difference in trends. If banking access improved faster in early-adopting states than in late-adopting states over these cohorts, the improvement would be attributed to the mandate. Identification therefore rests on an assumption about trends rather than levels.

Identification requires that, conditional on controls and fixed effects, treated and untreated cohorts would have experienced similar changes in banking participation in the absence of the mandate. I assess this assumption using event-study estimates, which compare outcomes across cohorts before and after mandate exposure. The coefficients on pre-treatment periods capture whether treated cohorts experienced differential changes in unbanked rates prior to the mandate. The event-study estimates in Figure 2 give no evidence of differential pre-treatment trends, and joint tests fail to reject the null of zero pre-treatment effects for women, men, and the female–male gap. The identifying assumption also requires no anticipation of treatment, which is plausible because the policy affects students through high school coursework and cannot directly change adult banking outcomes before exposure. Unlike Wooldridge (2025), who allows control variables to interact with each treatment cell and period, I include control variables additively, which trades some flexibility in the conditional parallel-trends assumption for greater precision.

A final consideration is that the two parts of the analysis use different sets of covariates, and this difference is deliberate. In the difference-in-differences design I control only for characteristics that are fixed before high school, namely race, ethnicity, and whether the respondent was born in the United States. I do not control for income, education, or employment, because the mandate can itself affect these outcomes, and conditioning on variables that the treatment can change would absorb part of the effect I am trying to measure. Adding them attenuates the estimated gender differential, as I report in appendix Table A4. I therefore study income, education, and employment as outcomes in Section 7.4, asking whether the mandate improves them, rather than holding them fixed here. This concern does not arise in the decomposition, which does not estimate the effect of the mandate at all, so there I include income, education, and employment alongside the other characteristics.

Column 4 of Table 6 is therefore my preferred specification. It is the one that estimates the difference between women's and men's effects, which is the parameter this study is about, and it conditions on the full set of pre-high-school characteristics while conditioning on nothing the mandate could itself have changed. Columns 2 and 3 reach it by adding those characteristics in steps, and I report the build-up because the estimated difference barely moves as they enter, which is itself evidence that the difference is not an artifact of composition.

7 Findings

In this section, I first present the decomposition results to characterize the sources of the gender gap in being unbanked. I then present the difference-in-differences estimates of the effect of state personal finance mandates, examine treatment effect heterogeneity, and present evidence on two potential mechanisms.

7.1 Decomposing the Gender Gap in Being Unbanked

In Table 4, I report the results of decomposing the 6.20 percentage point gender gap in unbanked status using the pooled linear probability model reported in Table 5. Observable characteristics explain 5.05 percentage points (81 percent) of the gap, leaving only 1.15 percentage points unexplained. Thus, most of the difference in banking status between unpartnered women and men reflects differences in their observed characteristics rather than differences in how those characteristics translate into banking outcomes.

The explained component is highly concentrated in three sets of characteristics. Family income accounts for 1.98 percentage points (32 percent of the total gap), household structure contributes 1.75 percentage points (28 percent), and race and ethnicity contribute another 1.06 percentage points (17 percent). Together, these three factors explain over three-quarters of the overall gender gap.

Income is the largest contributor because women are disproportionately represented at the lower end of the family income distribution. As shown in Table 5, only 15.0 percent of women report family income of at least $75,000 compared with 24.4 percent of men, while 21.7 percent of women—but only 13.1 percent of men—report family income below $15,000.

Since higher-income households are substantially less likely to be unbanked, these income differences alone account for nearly one-third of the overall gender gap.

Household structure is the second largest contributor. Nearly all of this contribution reflects the much higher prevalence of never-married mothers among women (28.1 percent versus 5.4 percent among men), a group with substantially higher predicted unbanked rates than never-married adults without children. Race and ethnicity also contribute meaningfully, primarily because women are more likely than men to be Black, while Black respondents have considerably higher predicted probabilities of being unbanked.

Differences in labor force participation explain a comparatively modest 7 percent of the gap because women are somewhat more likely to be unemployed or out of the labor force. In contrast, age and nativity contribute almost nothing because women and men in the sample differ little along either dimension.

Education is the only characteristic that narrows rather than widens the gender gap. Women are more educated than men in the sample, making them less likely to possess characteristics associated with being unbanked. If women instead had the same educational distribution as men, the gender gap would be slightly larger. Thus, educational attainment offsets part of the disadvantage associated with women's lower incomes and different household circumstances.

One characteristic absent from the explained component is exposure to state personal finance mandates. Adding mandate exposure to the decomposition does not change the results because women and men are equally likely to have been exposed. Decompositions explain gaps using characteristics that differ across groups, whereas policy exposure is assigned by state and graduation cohort rather than by gender. Whether these mandates reduce unbanked status is therefore a causal question, which I examine in Section 7.2 using the staggered adoption of state mandates.

Overall, the gender gap in being unbanked is largely attributable to differences in women's socioeconomic characteristics—especially lower family income, greater prevalence of single parenthood, and racial and ethnic composition—while women's higher educational attainment partially offsets these disadvantages.

7.2 Mandates Reduce the Gender Gap in Being Unbanked

In Table 6, I report the ETWFE estimates of the effect of high school personal finance mandates on the probability of being unbanked among unpartnered young adults. Across specifications, mandate exposure is associated with a meaningful reduction in the likelihood of being unbanked. In the simplest specification (column 1), mandate exposure is predicted to reduce the probability of being unbanked by 3.06 percentage points, equivalent to roughly 30 percent of the 10.28 percent baseline unbanked rate.

The average treatment effect, however, masks substantial gender heterogeneity. Columns 2 through 4 allow the effect of mandate exposure to differ between women and men by interacting the treatment indicator with an indicator for female. In the preferred specification (column 4), mandate exposure is predicted to reduce men's probability of being unbanked by 1.78 percentage points. Women experience an additional reduction of 2.66 percentage points, implying a total decline of 4.44 percentage points. Thus, the estimated effect for women is about two and a half times as large as that for men. Relative to the baseline rates reported in the first row of Table 1, the mandate is predicted to reduce men's unbanked rate by 25.3 percent (1.78/7.04), women's by 33.5 percent (4.44/13.24), and the female–male gap by 42.9 percent (2.66/6.20). Taken together, these estimates indicate that high school personal finance mandates substantially narrow the gender gap in being unbanked.

This result is robust to the inclusion of demographic controls. The estimated female differential changes little as controls for race, ethnicity, and nativity are added across columns 2 through 4, suggesting that the larger effect for women is not driven by compositional differences between women and men.

In Figure 2, I present event-study estimates of the effect of mandate exposure by years relative to the first graduation cohort exposed to a state's personal finance requirement, separately for women and men.[8] The event-study serves two purposes. First, the pre-treatment estimates are close to zero for both groups, providing no evidence of differential trends before the mandate-exposed cohorts enter the sample. Second, the decline in being unbanked emerges only after the first exposed cohort and is concentrated among women. For women, the estimated effects become negative immediately following exposure, with the first two post-treatment coefficients statistically distinguishable from zero. Although the estimates become less precise at longer horizons because fewer treated cohorts contribute to the estimation, they remain consistently negative. In contrast, the post-treatment estimates for men remain small and statistically indistinguishable from zero throughout. Overall, the event-study supports the identifying assumption underlying the difference-in-differences design and reinforces the main finding that personal finance mandates primarily reduce unbanked status among women.

7.3 Which Women Benefit Most from the Mandates?

The estimates in columns 1 through 4 of Table 6 summarize the average effects of mandate exposure in the full sample, allowing the effect to differ by gender. I next examine whether the additional effect for women differs across groups that may have faced different opportunities to benefit from financial education during high school. If personal finance mandates are particularly valuable for women who would otherwise have fewer opportunities to receive financial education, their effects should be largest among groups that were relatively disadvantaged while in high school. To examine this prediction, I re-estimate Equation (1) separately for three subsamples defined by characteristics that plausibly proxy for disadvantage during high school: Black or Hispanic respondents, foreign-born respondents, and respondents with no education beyond high school.[9]

[8] The recent difference-in-differences literature proposes several estimators for staggered treatment adoption. As a robustness check, I report event-study estimates obtained using ETWFE, Callaway–Sant'Anna, Sun–Abraham, and dynamic TWFE in Figure A1.

[9] Students from low-income households are another natural group to examine, since they may have the most to gain from a high-school personal finance course. I do not split the sample by income, however,

The evidence is strongest among respondents with no education beyond high school (column 7). For this group, mandate exposure reduces women's probability of being unbanked by an additional 7.63 percentage points relative to men, nearly three times the full-sample estimate reported in column 4. This subgroup also has the highest baseline unbanked rate (22.66 percent), and because high school represents their last formal educational experience, they are the group most likely to rely on the mandate as their primary source of personal finance education.

A similar, though smaller, pattern emerges among Black and Hispanic respondents. Their baseline unbanked rate (18.56 percent) is substantially higher than the full-sample average of 10.28 percent, and the estimated additional effect for women is 4.10 percentage points, larger than the full-sample estimate. By contrast, the estimated differential effect among foreign-born respondents is small and statistically insignificant. This subgroup has the lowest baseline unbanked rate (9.65 percent), and the female–male gap in unbanked stats itself is small (see Table 1), leaving relatively little gender gap for the mandate to reduce.

Taken together, these results suggest that the additional benefits of high-school personal finance mandates for women are largest among groups that appear to have had fewer opportunities to acquire financial knowledge outside the required curriculum. The strongest evidence comes from respondents whose education ended with high school, for whom the estimated additional effect for women is 7.63 percentage points. Relative to this group's baseline female–male difference in unbanked status of 12.14 percentage points (Table 1), this estimate implies that the mandates would reduce the gender gap by approximately 62.85 percent. Black and Hispanic respondents exhibit a similar, though smaller, pattern. The estimated additional effect of 4.10 percentage points amounts to a 54.23 percent reduction

because the survey measures family income at the time of the interview, when respondents are between ages 22 and 36, rather than during high school when mandate exposure occurred. Moreover, as shown in Section 7.4, mandate exposure itself affects women's income distribution, making current income a post-treatment outcome rather than an appropriate proxy for disadvantage during high school. Educational attainment is not a perfect measure either, but by these ages it largely reflects whether respondents were able to pursue education beyond high school, making it an informative proxy for disadvantage during high school.

relative to the subgroup's baseline gender gap of 7.56 percentage points.

7.4 Mechanisms: Why Women Experience Larger Effects Than M the Mandates Help Women More

Several mechanisms could explain why the mandates reduce unbanked rates more for women than for men, and I examine two that the available data allow me to test. First, the mandates may improve women's financial knowledge more than men's, increasing their ability and confidence to use formal financial services. Second, the mandates may improve economic outcomes, such as educational attainment, employment, and income, thereby changing the circumstances in which banking decisions are made. These improvements can both increase the benefits of holding a bank account and reduce the barriers to opening one. I examine the second mechanism using the FDIC sample, which contains information on education, employment, and income, and the first using the NFCS sample described in Section 4, because the FDIC survey does not measure financial knowledge.

7.4.1 Economic and Human Capital Outcomes

To examine whether the larger reduction in women's unbanked rates is accompanied by improvements in education and economic outcomes that could plausibly mediate the effect, I re-estimate Equation (1) using alternative outcomes in place of the unbanked indicator while leaving the specification otherwise unchanged. I consider two education outcomes: college completion, defined as holding a bachelor's degree or higher, and some college, which includes respondents who attended a two-year or four-year college without completing a bachelor's degree. I also examine labor force status (employed, unemployed, and out of the labor force), self-employment separately for incorporated and unincorporated businesses, and four mutually exclusive family-income categories. Because the sample is restricted to unpartnered adults, family income primarily reflects respondents' own economic resources, although it may also include the income of other related household members.

The education results suggest that the mandates increase women's likelihood of earning a bachelor's degree, with little evidence that they simply increase attendance without completion. There is essentially no differential effect on attending some college without earning a bachelor's degree, with an estimated female interaction of -0.78 percentage points on a mean of 33.19 percent. By contrast, the mandates are predicted to increase men's probability of completing college by 0.90 percentage points and women's by an additional 2.92 percentage points, implying a 3.82 percentage point increase for women, or about 9 percent of the 41.51 percent mean. Although the estimate for men is not statistically distinguishable from zero, the female differential is precisely estimated, suggesting that the educational gains are concentrated among women.

The mandates do not appear to change labor force attachment. The estimated female–male differences in employment, unemployment, and labor force participation are all small and statistically insignificant, and the estimates are sufficiently informative to rule out economically meaningful changes. By contrast, among employed respondents the mandates do appear to affect the type of work women perform. Exposure increases men's probability of self-employment by 1.31 percentage points and women's by a further 1.56 percentage points, implying an increase of 2.87 percentage points for women—more than half of the 5.08 percent baseline self-employment rate. Nearly all of this additional increase comes from unincorporated self-employment. The female interaction is 1.27 percentage points on a mean of 3.56 percent, while the corresponding interaction for incorporated self-employment is small and statistically insignificant. Thus, the mandates appear to encourage women to enter self-employment primarily through unincorporated businesses.

The income results suggest that the mandates raise family income for both men and women, although the gains occur at different points in the income distribution. Men experience the largest increase in the highest income category ($75,000 or more), with an estimated gain of 3.54 percentage points. Women's gains, in contrast, are concentrated in the middle of the distribution. The female interaction is 3.93 percentage points for the $50,000–$75,000 category, while it is −2.40 percentage points for the highest income category, implying that women's estimated increase at \$75,000 or more is 1.14 percentage points. Thus, the mandates appear to improve women's economic position primarily by moving more women into middle-income households rather than into the highest income category.

The results therefore suggest that the mandates improve women's human capital and economic circumstances along several dimensions while leaving labor force attachment largely unchanged. The strongest evidence is for increases in college completion, unincorporated self-employment, and movement into the middle of the income distribution. Each of these changes plausibly increases both the benefits of having a bank account and the resources available to maintain one. Although these reduced-form estimates cannot identify how much of the reduction in the gender gap in unbanked status operates through these channels, they are consistent with improvements in women's human capital and economic opportunities being one mechanism through which the mandates disproportionately benefit women.

7.4.2 Financial Knowledge

I conclude the analysis by examining whether the mandates improve financial knowledge, the outcome they are designed to affect directly. Because the FDIC survey contains no information on financial education or financial knowledge, I instead use the NFCS, which records both whether respondents received financial education and what they know, allowing me to test the policy's intended mechanism directly.

As described in Section 4, the NFCS sample consists of unpartnered high school graduates ages 18–44. Because the survey reports age only in broad bands rather than exact birth years, graduation cohorts can be identified only approximately. I therefore estimate conventional TWFE models with state, graduation-cohort, and survey-year fixed effects, interacting mandate exposure with the female indicator and clustering standard errors at the state level.

Table 8 reports the results. Columns 1–3 examine whether mandate exposure increases reported financial education, while columns 4–10 examine whether those increases translate into greater financial knowledge.

Columns 1–3 show that the mandates increase reported exposure to financial education. Mandate exposure raises the probability that respondents report receiving any financial education and financial education specifically in high school. The estimated female interaction terms are small and statistically insignificant throughout. The mandates therefore appear to increase access to financial education similarly across genders.

Columns 4–10 then examine whether this additional exposure translates into greater financial knowledge. For men, there is little evidence that it does. The estimated mandate effects are small and statistically insignificant across all knowledge measures. Women, however, exhibit larger gains. On the five-question financial knowledge score in column 4, mandate exposure increases women's score by an additional 0.13 points relative to men. Because the estimated effect for men is small and negative, this corresponds to a net increase of 0.08 points for women. The largest improvements occur on the two investment questions. Women become 4.35 percentage points more likely than men to answer the bond-price question correctly (column 7) and 4.28 percentage points more likely to correctly identify that a single stock is riskier than a stock mutual fund (column 9), representing roughly one-fifth and one-eighth of the respective baseline rates. The differential effect on the compound-interest question is smaller, at 2.61 percentage points (column 5). By contrast, self-assessed financial knowledge does not change for either gender (column 10).

Taken together, these results suggest that the larger effects of the mandates for women do not arise because women are more likely than men to receive financial education. Rather, women appear to acquire more financial knowledge from the same increase in educational exposure. Although the NFCS cannot identify why the same coursework generates larger knowledge gains for women, the pattern is consistent with the larger reductions in women's unbanked rates documented earlier and supports the interpretation that improvements in financial knowledge are one channel through which the mandates reduce gender disparities.

8 Conclusion

Among unpartnered young adults, whose banking status reflects their own decisions rather than a partner's, women are substantially more likely than men to be unbanked. In this study, I ask whether high school personal finance mandates reduce that gap. Using the FDIC National Survey of Unbanked and Underbanked Households and a heterogeneity robust difference-in-differences design, I find that exposure to the mandate reduces the probability of being unbanked by 3.1 percentage points overall, equivalent to roughly 30 percent of the baseline unbanked rate. The effects are substantially larger for women than for men. Women experience a reduction about two and a half times as large, closing 42.9 percent of the gender gap in unbanked status among unpartnered young adults. The largest gains occur among women with the lowest baseline banking participation rates, particularly those with no education beyond high school and Black and Hispanic women.

The evidence points to both financial knowledge and subsequent economic outcomes as plausible mechanisms for why the mandates have larger effects for women than for men. Using NFCS data, I find that mandates increase women's measured financial knowledge more than men's, with the largest improvements occurring on questions about investment concepts while leaving women's self assessed financial knowledge unchanged. The FDIC evidence also indicates improvements in subsequent economic circumstances. For women, the mandates are associated with higher rates of college completion, unincorporated self employment, and movement into the middle of the income distribution.

These findings contribute to the broader discussion of how to improve financial inclusion. Many adult financial education programs (Fernandes, Lynch and Netemeyer, 2014) and policies that expand access to bank accounts (Dupas et al., 2018; Collins, Larrimore and Urban, 2026) have produced limited effects. One reason may be that voluntary financial education primarily reaches individuals who are already engaged with the financial system, whereas high school personal finance mandates reach every student in an affected cohort before they begin making independent financial decisions. The FDIC data also provide little evidence that lack of interest in banking is the primary barrier for these women. Unbanked women express a greater desire to open an account than unbanked men. Together, these findings imply that financial education and account access policies are complements rather than substitutes. Expanding the availability of low cost accounts can reduce supply side barriers, but equipping young adults with financial knowledge before they enter adulthood addresses informational barriers that access alone cannot overcome. Doing so can play an important role in improving financial inclusion and reducing gender disparities.

This study also has several limitations that point to directions for future research. Because the FDIC records banking status at the household level, the analysis is restricted to unpartnered young adults whose banking status reflects their own decisions. Richer data that measure banking participation at the individual level within households would make it possible to examine how personal finance mandates affect joint banking decisions among partnered adults. In addition, while the mechanism analyses indicate that mandates improve both financial knowledge and subsequent economic outcomes, they rely on different surveys and therefore cannot identify the relative importance of each channel. Richer longitudinal data that jointly measure financial knowledge, economic outcomes, and banking behavior would help disentangle these mechanisms. Despite these limitations, the results indicate that high school personal finance mandates can play an important role in improving financial inclusion and narrowing gender disparities early in adulthood.

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Tables

Table 1: Unbanked rates by gender and respondent characteristics

	Female	Male	Difference	N (F)	N (M)
Characteristic	(1)	(2)	(3)	(4)	(5)
All unpartnered respondents	13.24 (0.66)	7.04 (0.50)	+6.20***	13,213	11,742
Household structure					
never married, with children	26.19 (1.09)	15.52 (2.14)	+10.67***	3,480	579
never married, no children	5.70 (0.43)	6.09 (0.47)	−0.39	6,612	9,469
formerly married, with children	15.14 (0.94)	8.55 (1.47)	+6.59***	2,008	367
formerly married, no children	9.22 (1.42)	9.77 (1.44)	−0.55	696	1,036
married (spouse absent), with children	19.57 (3.00)	7.72 (4.49)	+11.85***	254	27
married (spouse absent), no children	13.42 (4.15)	9.06 (1.48)	+4.36	163	264
Race and ethnicity					
White	6.81 (0.62)	3.74 (0.29)	+3.06***	7,445	7,874
Black	25.44 (1.39)	17.00 (1.27)	+8.44***	2,784	1,238
Hispanic	15.42 (1.60)	11.12 (1.04)	+4.30***	1,805	1,403
other race or ethnicity	7.71 (1.70)	3.73 (0.66)	+3.98**	1,179	1,227
Family income					
under \$15,000	35.13 (1.26)	23.87 (1.71)	+11.26***	2,961	1,540
\$15,000 to \$30,000	17.15 (0.98)	12.32 (1.08)	+4.83***	3,028	1,962
\$30,000 to \$50,000	4.63 (0.44)	4.59 (0.44)	+0.05	3,246	3,016
\$50,000 to \$75,000	2.77 (0.44)	2.29 (0.36)	+0.48	2,085	2,503
\$75,000 or more	1.39 (0.36)	1.18 (0.34)	+0.20	1,893	2,721
Education					
high school graduate	28.84 (1.24)	16.69 (1.21)	+12.14***	3,134	3,157
some college	15.61 (1.00)	6.73 (0.62)	+8.87***	4,578	3,645
college or more	2.15 (0.32)	1.10 (0.14)	+1.05***	5,501	4,940
Labor force participation					
employed	8.40 (0.56)	5.03 (0.45)	+3.38***	10,698	10,091
unemployed	35.25 (2.27)	21.34 (2.61)	+13.91***	844	605
not in the labor force	32.29 (1.29)	16.95 (1.71)	+15.33***	1,671	1,046
Foreign-born (born outside the U.S.)	10.14 (1.05)	9.23 (1.22)	+0.91	1,350	1,472
Age					
22 to 26	12.45 (0.89)	6.58 (0.60)	+5.87***	4,064	3,839
27 to 31	13.61 (0.87)	6.65 (0.61)	+6.95***	4,738	4,253
32 to 36	13.63 (0.78)	8.05 (0.68)	+5.58***	4,411	3,650

Note: I report the survey-weighted share unbanked, by gender, in the unpartnered sample. Household structure combines marital history with the presence of a co-resident child aged 18 or younger; married respondents are spouse-absent, since respondents with a co-resident spouse or partner are excluded by construction, and formerly married covers divorced, separated, and widowed respondents. The Difference column is the female share minus the male share, in percentage points; the spouse-absent rows contain few male respondents, so their differences are imprecisely estimated. The foreign-born row is the share unbanked among respondents born outside the United States, to be read against the whole-sample row at the top; labor force participation and family income are measured at the survey date rather than at high school age. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t-test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Sample means of banking status and covariates by gender

	Total	Female	Male	Difference
Variable	(1)	(2)	(3)	(4)
Banking status				
1 if unbanked	10.28 (0.54)	13.24 (0.66)	7.04 (0.50)	+6.20***
Household structure				
1 if never married, with children	17.24 (0.55)	28.10 (0.92)	5.38 (0.49)	+22.72***
1 if never married, no children	65.09 (0.95)	50.47 (1.25)	81.06 (0.73)	−30.59***
1 if formerly married, with children	8.34 (0.44)	13.50 (0.70)	2.71 (0.23)	+10.79***
1 if formerly married, no children	6.54 (0.30)	5.02 (0.28)	8.19 (0.50)	−3.17***
1 if married (spouse absent), with children	1.07 (0.06)	1.81 (0.11)	0.25 (0.05)	+1.56***
1 if married (spouse absent), no children	1.73 (0.15)	1.10 (0.14)	2.41 (0.19)	−1.31***
Race and ethnicity				
1 if White	52.06 (2.65)	47.94 (2.37)	56.56 (3.07)	−8.63***
1 if Black	21.52 (2.21)	26.56 (2.64)	16.02 (1.70)	+10.54***
1 if Hispanic	16.14 (2.74)	16.32 (2.81)	15.95 (2.70)	+0.37
1 if other race or ethnicity	10.27 (1.32)	9.19 (1.18)	11.46 (1.51)	−2.28***
Family income				
1 if family income is under \$15,000	17.61 (0.70)	21.74 (0.97)	13.10 (0.51)	+8.65***
1 if family income is \$15,000 to \$30,000	19.35 (0.75)	22.20 (0.75)	16.23 (0.89)	+5.97***
1 if family income is \$30,000 to \$50,000	24.55 (0.68)	24.35 (0.67)	24.77 (0.82)	−0.42
1 if family income is \$50,000 to \$75,000	18.99 (0.42)	16.71 (0.51)	21.48 (0.54)	−4.77***
1 if family income is \$75,000 or more	19.51 (1.63)	15.00 (1.45)	24.43 (1.86)	−9.42***
Education				
1 if high school graduate	25.30 (0.96)	23.79 (0.94)	26.94 (1.19)	−3.15***
1 if some college	33.19 (0.98)	35.27 (1.17)	30.91 (0.96)	+4.36***
1 if college or more	41.51 (1.55)	40.94 (1.71)	42.15 (1.57)	−1.21
Labor force participation				
1 if employed	82.69 (0.58)	80.56 (0.64)	85.02 (0.67)	−4.46***
1 if unemployed	6.00 (0.27)	6.67 (0.31)	5.26 (0.31)	+1.40***
1 if not in the labor force	11.31 (0.40)	12.77 (0.48)	9.72 (0.48)	+3.06***
Foreign-born (1 if born outside the U.S.)	13.47 (1.51)	11.78 (1.51)	15.32 (1.53)	−3.54***
Age in years	28.90 (0.05)	29.02 (0.06)	28.77 (0.06)	+0.25***
Observations	24,955	13,213	11,742	

Note: I report survey-weighted means of respondent characteristics, by gender. Household structure combines partnership status with the presence of a co-resident child aged 18 or younger; married respondents are spouse-absent, since respondents with a co-resident spouse or partner are excluded by construction, and formerly married covers divorced, separated, and widowed respondents. The Difference column is the female mean minus the male mean. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t-test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Reasons for being unbanked, by gender (2013–2023)

	Total	Female	Male	Difference
Reason	(1)	(2)	(3)	(4)
Don't have enough money to meet minimum balance	49.92	52.20	45.26	+6.93**
Don't trust banks	34.99	32.72	39.64	−6.92**
Bank account fees too high or unpredictable	34.24	32.88	37.01	−4.13
Avoiding a bank gives more privacy	30.47	29.76	31.93	−2.17
Don't have required personal identification	15.53	14.59	17.45	−2.86
Banks don't offer needed products or services	15.25	14.93	15.91	−0.98
Inconvenient bank hours or locations	12.27	11.21	14.44	−3.23
Observations (unbanked)	1,559	1,072	487	

Note: I report the survey-weighted share of unbanked respondents citing each reason, by gender. Respondents can report more than one reason, so the shares do not sum to 100. The Difference column is the female share minus the male share, in percentage points. Stars are from a weighted, state-clustered t-test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Kitagawa–Oaxaca–Blinder decomposition of the gender gap in unbanked status

	Contribution	Share
	(1)	(2)
Total gap (Female − Male)	**+6.20**	**100%**
	(0.63)	
Explained	**+5.05**	**+81%**
	(0.42)	
Family income	+1.98	+32%
	(0.15)	
Household structure	+1.75	+28%
	(0.22)	
Race and ethnicity	+1.06	+17%
	(0.12)	
Labor force participation	+0.42	+7%
	(0.07)	
Age	+0.06	+1%
	(0.01)	
Foreign-born	+0.01	+0%
	(0.02)	
Education	−0.23	−4%
	(0.10)	
Unexplained	**+1.15**	**+19%**
	(0.40)	

Note: The decomposition is based on a linear probability model; the explained component prices the female−male difference in characteristics with coefficients from the pooled sample. Contributions are in percentage points and block contributions sum to the explained total by construction; column 2 gives each row's contribution as a share of the total gap. The sample is 24,955 respondents (13,213 women and 11,742 men), and the underlying means and coefficients are in Table 5. Standard errors, below each estimate, are from 400 state-clustered bootstrap resamples.

Table 5: Sample means and estimated coefficients underlying the decomposition in Table 4

	Sample mean		Difference	Estimated	Contribution
	Women	Men	(Women − Men)	coefficient	to the gap
	(1)	(2)	(3)	(4)	(5)
Family income					**+1.98 (0.15)**
1 if family income is under $15,000 (reference category)	0.217 (0.010)	0.131 (0.005)	+0.086 (0.008)		
1 if family income is $15,000 to $30,000	0.222 (0.007)	0.162 (0.008)	+0.060 (0.007)	−12.03 (1.06)	−0.72 (0.10)
1 if family income is $30,000 to $50,000	0.243 (0.006)	0.248 (0.008)	−0.004 (0.006)	−19.33 (0.93)	+0.08 (0.12)
1 if family income is $50,000 to $75,000	0.167 (0.005)	0.215 (0.006)	−0.048 (0.006)	−18.70 (0.95)	+0.89 (0.12)
1 if family income is $75,000 or more	0.150 (0.014)	0.244 (0.018)	−0.094 (0.008)	−18.32 (0.86)	+1.73 (0.15)
Household structure					**+1.75 (0.22)**
1 if never married, no children (reference category)	0.505 (0.012)	0.811 (0.007)	−0.306 (0.010)		
1 if never married, with children	0.281 (0.009)	0.054 (0.005)	+0.227 (0.011)	+7.10 (0.78)	+1.61 (0.22)
1 if formerly married, with children	0.135 (0.007)	0.027 (0.002)	+0.108 (0.006)	+1.79 (0.82)	+0.19 (0.09)
1 if formerly married, no children	0.050 (0.003)	0.082 (0.005)	−0.032 (0.005)	+1.92 (0.85)	−0.06 (0.03)
1 if married (spouse absent), with children	0.018 (0.001)	0.003 (0.001)	+0.016 (0.001)	+3.43 (2.60)	+0.05 (0.04)
1 if married (spouse absent), no children	0.011 (0.001)	0.024 (0.002)	−0.013 (0.002)	+3.65 (1.46)	−0.05 (0.02)
Race and ethnicity					**+1.06 (0.12)**
1 if White (reference category)	0.479 (0.024)	0.566 (0.031)	−0.086 (0.012)		
1 if Black	0.266 (0.025)	0.160 (0.016)	+0.105 (0.012)	+10.08 (0.90)	+1.06 (0.13)
1 if Hispanic	0.163 (0.028)	0.160 (0.027)	+0.004 (0.006)	+4.11 (0.73)	+0.02 (0.03)
1 if other race or ethnicity	0.092 (0.011)	0.115 (0.014)	−0.023 (0.006)	+0.96 (0.71)	−0.02 (0.01)
Labor force participation					**+0.42 (0.07)**
1 if employed (reference category)	0.806 (0.006)	0.850 (0.007)	−0.045 (0.006)		
1 if unemployed	0.067 (0.003)	0.053 (0.003)	+0.014 (0.003)	+11.42 (1.26)	+0.16 (0.04)
1 if not in the labor force	0.128 (0.005)	0.097 (0.005)	+0.031 (0.005)	+8.59 (0.75)	+0.26 (0.05)
Age (in years)	29.019 (0.064)	28.767 (0.056)	+0.252 (0.050)	+0.22 (0.05)	**+0.06 (0.01)**
Foreign-born (1 if born outside the U.S.)	0.118 (0.014)	0.153 (0.015)	−0.035 (0.004)	−0.32 (0.64)	**+0.01 (0.02)**
Education					**−0.23 (0.10)**
1 if high school graduate	0.238 (0.009)	0.269 (0.012)	−0.031 (0.008)	+11.49 (0.81)	−0.36 (0.11)
1 if some college	0.353 (0.011)	0.309 (0.009)	+0.044 (0.008)	+2.98 (0.47)	+0.13 (0.03)
1 if college or more (reference category)	0.409 (0.016)	0.421 (0.015)	−0.012 (0.010)		

Note: I report the components underlying the decomposition in Table 4. Column 4 gives coefficients, in percentage points, from a linear probability model of unbanked status fit to the pooled sample with an indicator for gender (Fortin, Lemieux and Firpo, 2011). Each categorical coefficient is relative to the block's reference category, shown as its own row with columns 4 and 5 blank; age enters in years. Column 5 is each characteristic's contribution to the female−male gap, the product of columns 3 and 4 up to rounding, and within each block the contributions sum to the bold block total, which matches that block's row in Table 4. Refitting the pooled model without the gender indicator changes the explained share from 81 percent to 85 percent. The sample is 24,955 respondents (13,213 women and 11,742 men). Standard errors in parentheses are from 400 state-clustered bootstrap resamples.

Table 6: ETWFE estimates of the effect of high school personal finance mandates on the probability of being unbanked

	Dependent variable: 1 if unbanked						
	Full sample				Subgroups		
					Black/Hisp	Foreign born	HS only
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mandate exposure × Female	–	−2.78***	−2.68***	−2.66***	−4.10**	0.19	−7.63***
		(0.80)	(0.69)	(0.69)	(1.85)	(1.91)	(1.77)
Mandate exposure	−3.06***	−1.76**	−1.78**	−1.78**	−4.63**	−6.77***	−1.64
	(0.86)	(0.87)	(0.76)	(0.76)	(1.87)	(2.19)	(1.66)
1 if female	–	6.72***	4.76***	4.71***	7.25***	−1.00	10.48***
		(0.63)	(0.52)	(0.52)	(1.10)	(1.03)	(1.37)
1 if Black	–	–	17.80***	17.89***	8.11***	3.95*	25.22***
			(1.19)	(1.18)	(1.52)	(2.03)	(1.81)
1 if Hispanic	–	–	9.56***	9.95***	–	18.04***	12.56***
			(0.94)	(0.93)		(2.07)	(1.58)
1 if other race or ethnicity	–	–	2.56**	3.12***	–	−0.57	9.17***
			(1.10)	(1.20)		(1.26)	(2.68)
1 if foreign-born	–	–	–	−1.53**	1.51	–	1.86
				(0.66)	(1.00)		(1.95)
Mean dep. var. (%)	10.28	10.28	10.28	10.28	18.56	9.65	22.66
Observations	24,955	24,955	24,955	24,955	7,230	2,822	6,291
Adjusted R^2	0.007	0.018	0.066	0.066	0.029	0.097	0.095

Note: All columns are ETWFE estimates with adoption-cohort and graduation-cohort fixed effects, and coefficients are in percentage points. Columns 5–7 restrict the sample to Black or Hispanic respondents, foreign-born respondents, and respondents with no education beyond high school. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: ETWFE estimates of the effect of high school personal finance mandates on human capital and economic outcomes

	Education		Labor force status			Self-employment among employed			Income			
	Some college	College+	Employed	Unemployed	Not in LF	Self-emp.	SE inc.	SE uninc.	< 30k	30-50k	50-75k	75k+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Mandate exposure × Female	−0.78	2.92***	1.22	0.04	−1.26	1.56***	0.29	1.27***	−0.78	−0.75	3.93***	−2.40**
	(1.04)	(1.07)	(1.31)	(0.43)	(1.05)	(0.49)	(0.29)	(0.44)	(1.52)	(1.04)	(1.03)	(1.02)
Mandate exposure	−0.14	0.90	−0.57	0.48	0.10	1.31**	1.08***	0.23	−1.80	−1.58	−0.17	3.54***
	(1.65)	(2.27)	(1.18)	(0.61)	(1.11)	(0.65)	(0.36)	(0.56)	(1.33)	(1.29)	(1.45)	(1.34)
1 if female	2.63***	1.94**	−4.15***	0.43	3.72***	−3.16***	−1.43***	−1.73***	13.66***	−1.04	−5.62***	−7.01***
	(0.80)	(0.88)	(0.69)	(0.29)	(0.58)	(0.28)	(0.15)	(0.24)	(1.19)	(0.75)	(0.82)	(0.72)
1 if Black	6.97***	−19.77***	−10.23***	5.92***	4.32***	−1.19***	−0.46**	−0.74*	16.43***	−2.44**	−5.91***	−8.08***
	(1.18)	(1.46)	(1.04)	(0.49)	(0.75)	(0.42)	(0.22)	(0.39)	(1.53)	(0.98)	(0.86)	(1.15)
1 if Hispanic	9.19***	−24.62***	−2.53**	1.92***	0.62	−0.59	−0.38	−0.21	8.77***	0.54	−3.13***	−6.17***
	(1.48)	(1.70)	(1.00)	(0.47)	(0.75)	(0.61)	(0.23)	(0.50)	(1.49)	(0.98)	(1.03)	(1.73)
1 if other race or ethnicity	−0.89	4.74	−6.80***	1.28*	5.52***	−0.27	−0.44	0.18	2.05	−6.34***	−2.19**	6.49***
	(1.73)	(3.07)	(1.37)	(0.71)	(1.03)	(0.65)	(0.30)	(0.56)	(1.97)	(1.11)	(0.86)	(1.92)
1 if foreign-born	−10.65***	13.70***	0.46	−2.30***	1.85**	0.39	−0.01	0.40	−1.00	−2.16**	0.64	2.52***
	(0.94)	(1.99)	(1.05)	(0.54)	(0.78)	(0.85)	(0.27)	(0.76)	(1.27)	(1.04)	(1.24)	(0.87)
Mean dep. var. (%)	33.19	41.51	82.69	6.00	11.31	5.08	1.52	3.56	36.96	24.55	18.99	19.51
Observations	24,955	24,955	24,955	24,955	24,955	20,789	20,789	20,789	24,955	24,955	24,955	24,955
Adjusted R^2	0.019	0.062	0.018	0.012	0.011	0.007	0.003	0.003	0.049	0.006	0.009	0.044

Note: Each column is a separate ETWFE regression of a binary outcome, with adoption-cohort and graduation-cohort fixed effects, and coefficients are in percentage points. Self-employment outcomes are measured among employed respondents only. The income categories are mutually exclusive: under \$30k, \$30–50k, \$50–75k, and \$75k or more. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

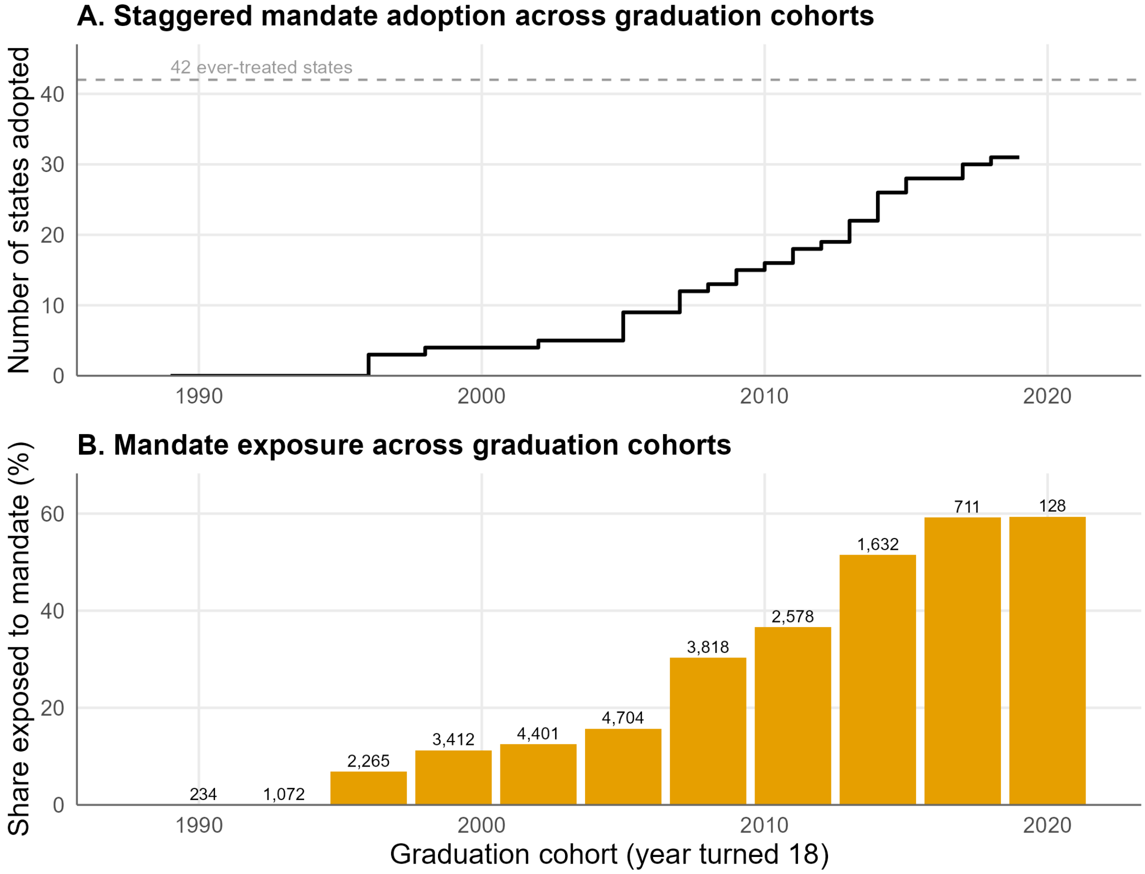
Table 8: TWFE estimates of the effect of high school personal finance mandates on financial education and financial knowledge

	Financial education			Financial knowledge						Self-rated
	Offered	Received	Received in HS	Score (0–5)	Comp. interest	Inflation	Bonds	Mortgage	Stocks	Score (1–7)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Mandate exposure × Female	0.54	0.47	0.29	0.13***	2.61*	1.73	4.35***	0.13	4.28***	−0.03
	(1.26)	(0.92)	(0.74)	(0.04)	(1.49)	(1.59)	(0.81)	(1.04)	(1.30)	(0.04)
Mandate exposure	0.34	2.09*	2.57**	−0.05	−1.32	−0.26	−2.74***	0.35	−1.14	0.02
	(1.42)	(1.16)	(1.10)	(0.04)	(1.45)	(1.40)	(0.72)	(1.04)	(1.13)	(0.04)
1 if female	−9.10***	−4.57***	−3.97***	−0.40***	−5.93***	−11.67***	−7.08***	−3.26***	−12.11***	−0.43***
	(0.76)	(0.48)	(0.56)	(0.03)	(0.62)	(1.05)	(0.45)	(0.70)	(0.73)	(0.02)
1 if non-White	3.12***	2.12***	0.15	−0.25***	−5.28***	−7.57***	−0.53	−8.37***	−3.39***	0.09***
	(0.85)	(0.67)	(0.45)	(0.03)	(0.81)	(0.92)	(0.48)	(0.86)	(1.24)	(0.02)
Mean dep. var.	37.86	24.08	14.85	2.22	65.20	42.15	20.32	59.14	35.07	4.80
Observations	26,421	26,421	26,421	31,659	31,659	31,659	31,659	31,659	31,659	30,624
Adjusted R^2	0.037	0.022	0.030	0.076	0.032	0.051	0.011	0.042	0.040	0.049

Note: All columns are static TWFE estimates with state, graduation-cohort, and survey-year fixed effects, on the NFCS sample of unpartnered respondents. Columns 1–3 are indicators for financial education offered, received in school, college, or at work, and received in high school, the last available only in NFCS waves from 2012 onward. Column 4 is the number of the five Lusardi–Mitchell financial literacy questions a respondent answers correctly, from 0 to 5; columns 5–9 are those five questions taken one at a time, each an indicator equal to one for a correct answer. Column 10 is respondents' assessment of their own financial knowledge, on a 1–7 scale. Coefficients and mean dependent variables are in percentage points for the binary outcomes in columns 1–3 and 5–9, and in scale points for columns 4 and 10. Observations differ across columns because the outcomes are not asked in every wave. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

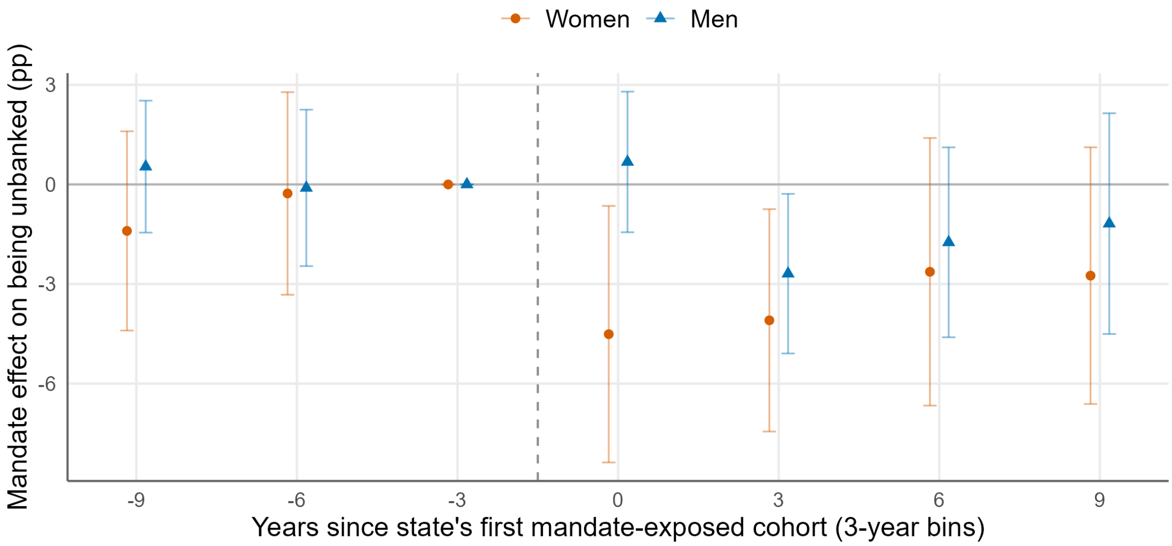
Figures

Figure 1: Staggered adoption of high school personal finance graduation requirements



Note: A graduation cohort is the year the respondent turned 18. Panel A shows the cumulative number of states whose requirement binds by each graduation cohort, and the dashed line marks the 42 ever-treated states. The count reaches 31 by the 2019 cohort because the remaining 11 states first bind on cohorts graduating in 2020 or later, outside the sample; they enter the estimation as not-yet-treated controls. Panel B shows the share of the analysis sample exposed to a mandate in each three-year graduation-cohort bin, with the number of observations above the bars.

Figure 2: Event study estimates of the effect of personal finance mandates on being unbanked by gender



Note: Estimates are from an additive extended two-way fixed-effects (ETWFE) specification with never-treated states as the comparison group, separately for women and men. Graduation cohorts are grouped into three-year bins, and estimates are normalized to the bin three years before the first exposed cohort. The horizontal axis is years since the state's first mandate-exposed cohort. Points are estimates in percentage points, and bars are 95 percent confidence intervals from state-clustered standard errors.

Appendix

Table A1: State personal finance mandates and first exposed graduating cohorts

State	First exposed cohort	Stand-alone course	In estimation sample
States with a personal finance graduation requirement (42)			
Illinois	1970	—	Yes
New Hampshire	1993	2027	Yes
New York	1996	—	Yes
Michigan	1998	2028	Yes
Wyoming	2002	—	Yes
Arizona	2005	—	Yes
Arkansas	2005	—	Yes
Louisiana	2005	2027	Yes
North Carolina	2005	2024	Yes
Georgia	2007	2028	Yes
Idaho	2007	—	Yes
Texas	2007	2030	Yes
Utah	2008	2008	Yes
Colorado	2009	2030	Yes
South Carolina	2009	2027	Yes
Missouri	2010	2010	Yes
Iowa	2011	2021	Yes
North Dakota	2011	—	Yes
Kansas	2012	2027	Yes
Indiana	2013	2028	Yes
Oregon	2013	2027	Yes
Tennessee	2013	2013	Yes
Nebraska	2014	2024	Yes
New Jersey	2014	—	Yes
Ohio	2014	2026	Yes
Oklahoma	2014	—	Yes
Minnesota	2015	2028	Yes
Virginia	2015	2015	Yes
Alabama	2017	2017	Yes
Maine	2017	—	Yes
Florida	2018	2027	Yes
California	2020	2031	No
West Virginia	2020	2028	No
Mississippi	2022	2022	No
Nevada	2023	—	No
Kentucky	2024	2030	No
Rhode Island	2024	2024	No
Connecticut	2027	2027	No
New Mexico	2027	—	No
Wisconsin	2028	2028	No
Delaware	2030	2030	No
Pennsylvania	2030	2030	No
States with no requirement (9)			
Alaska	—	—	—
District of Columbia	—	—	—
Hawaii	—	—	—
Maryland	—	—	—
Massachusetts	—	—	—
Montana	—	—	—
South Dakota	—	—	—
Vermont	—	—	—
Washington	—	—	—

Note: The first exposed cohort is the earliest high school graduating cohort required to complete personal finance instruction; a dash in the stand-alone column marks a state that meets the requirement through an embedded course instead. The last graduating cohort observed is 2019, so a state enters the estimation sample only if its first exposed cohort is that cohort or earlier. Adoption dates are from Urban et al. (2020) and Urban's compilation of state financial education requirements, last updated in 2024.

Table A2: Sample means of key variables by treatment status, FDIC survey sample

Variable	Total (1)	Ever-treated (2)	Never-treated (3)	Difference (4)
Banking status				
1 if unbanked	10.28 (0.54)	10.65 (0.56)	6.10 (0.94)	+4.54***
Household structure				
1 if never married, with children	17.24 (0.55)	17.39 (0.58)	15.48 (2.00)	+1.91
1 if never married, no children	65.09 (0.95)	64.65 (1.03)	70.07 (1.42)	−5.42***
1 if formerly married, with children	8.34 (0.44)	8.51 (0.48)	6.44 (0.54)	+2.07***
1 if formerly married, no children	6.54 (0.30)	6.64 (0.32)	5.32 (0.83)	+1.33
1 if married (spouse absent), with children	1.07 (0.06)	1.08 (0.06)	0.92 (0.14)	+0.16
1 if married (spouse absent), no children	1.73 (0.15)	1.72 (0.16)	1.77 (0.31)	−0.05
Race and ethnicity				
1 if White	52.06 (2.65)	51.84 (2.83)	54.54 (4.91)	−2.70
1 if Black	21.52 (2.21)	21.90 (2.34)	17.28 (7.38)	+4.61
1 if Hispanic	16.14 (2.74)	16.51 (2.94)	12.04 (1.74)	+4.46
1 if other race or ethnicity	10.27 (1.32)	9.76 (1.44)	16.13 (2.89)	−6.37**
Family income				
1 if family income is under $15,000	17.61 (0.70)	17.90 (0.78)	14.38 (0.56)	+3.52***
1 if family income is $15,000 to $30,000	19.35 (0.75)	19.64 (0.81)	16.01 (1.40)	+3.64**
1 if family income is $30,000 to $50,000	24.55 (0.68)	24.76 (0.75)	22.11 (1.12)	+2.66**
1 if family income is $50,000 to $75,000	18.99 (0.42)	18.97 (0.46)	19.21 (0.83)	−0.24
1 if family income is $75,000 or more	19.51 (1.63)	18.73 (1.80)	28.29 (1.94)	−9.57***
Education				
1 if high school graduate	25.30 (0.96)	25.66 (1.05)	21.17 (1.36)	+4.49***
1 if some college	33.19 (0.98)	33.54 (1.04)	29.19 (2.93)	+4.35
1 if college or more	41.51 (1.55)	40.80 (1.67)	49.64 (3.66)	−8.84**
Labor force participation				
1 if employed	82.69 (0.58)	82.50 (0.61)	84.92 (0.90)	−2.42**
1 if unemployed	6.00 (0.27)	6.10 (0.29)	4.87 (0.23)	+1.23***
1 if not in the labor force	11.31 (0.40)	11.41 (0.42)	10.22 (0.72)	+1.19
Foreign-born (1 if born outside the U.S.)	13.47 (1.51)	13.29 (1.65)	15.52 (1.57)	−2.24
Age in years	28.90 (0.05)	28.88 (0.06)	29.06 (0.15)	−0.18
Observations	24,955	20,843	4,112	

Note: I report survey-weighted means of respondent characteristics by treatment status. The Total column is the full analytic sample, and the Difference column is the ever-treated mean minus the never-treated mean. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t-test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Sample means of key variables by treatment status, NFCS sample

Variable	Total (1)	Ever-treated (2)	Never-treated (3)	Difference (4)
Financial knowledge				
Quiz score (0–5)	2.22 (0.02)	2.19 (0.02)	2.38 (0.04)	−0.19***
1 if compound interest correct	65.20 (0.74)	64.14 (0.66)	70.79 (1.05)	−6.65***
1 if inflation correct	42.15 (0.71)	41.27 (0.70)	46.73 (1.19)	−5.46***
1 if bond prices correct	20.32 (0.39)	20.39 (0.44)	19.92 (0.73)	+0.47
1 if mortgage interest correct	59.14 (0.69)	58.65 (0.74)	61.70 (1.39)	−3.05**
1 if stock-vs-fund risk correct	35.07 (0.59)	34.32 (0.52)	39.01 (1.61)	−4.68***
Self-assessed knowledge				
Self-rated knowledge (1–7)	4.80 (0.02)	4.81 (0.02)	4.74 (0.05)	+0.07
Financial education				
1 if offered financial education (school, college, or workplace)	37.86 (0.44)	37.77 (0.46)	38.36 (1.26)	−0.60
1 if received financial education (school, college, or workplace)	24.08 (0.50)	24.06 (0.52)	24.15 (1.63)	−0.09
1 if received financial education in high school	14.85 (0.39)	15.06 (0.43)	13.79 (0.99)	+1.27
Demographics and education				
1 if non-White	55.06 (3.16)	54.47 (3.54)	58.18 (7.21)	−3.71
1 if aged 18–24	38.98 (0.63)	39.93 (0.45)	34.00 (2.01)	+5.93***
1 if aged 25–34	34.89 (0.48)	34.46 (0.44)	37.15 (1.72)	−2.69
1 if aged 35–44	26.13 (0.49)	25.61 (0.44)	28.85 (1.42)	−3.24**
1 if high school graduate	34.18 (0.83)	35.32 (0.72)	28.24 (2.27)	+7.08***
1 if some college	41.90 (0.79)	42.37 (0.65)	39.49 (3.55)	+2.87
1 if college or more	23.91 (1.30)	22.32 (0.88)	32.27 (5.65)	−9.95*
Observations	31,659	26,218	5,441	

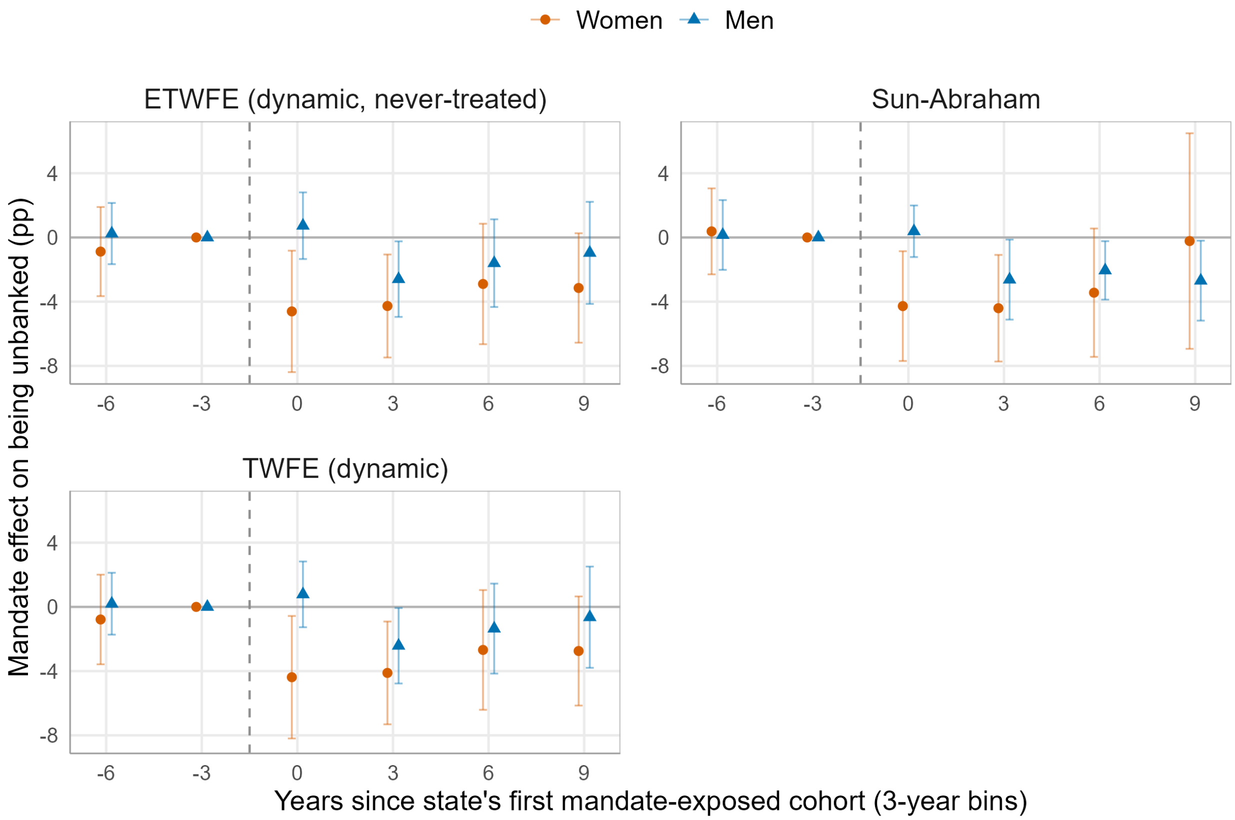
Note: I report survey-weighted means of NFCS respondent characteristics by treatment status. The Total column is the full analytic sample, and the Difference column is the ever-treated mean minus the never-treated mean. The financial education questions are available only in NFCS waves from 2012 onward. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t-test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: ETWFE estimates of the effect on the probability of being unbanked, adding post-treatment mediators as controls

	Dependent variable: 1 if unbanked				
	(1)	(2)	(3)	(4)	(5)
Mandate exposure × Female	−2.66***	−2.17***	−2.21***	−2.44***	−1.86***
	(0.69)	(0.57)	(0.70)	(0.59)	(0.51)
Mandate exposure	−1.78**	−1.53**	−1.63**	−1.89***	−1.64***
	(0.76)	(0.75)	(0.66)	(0.70)	(0.62)
1 if female	4.71***	2.04***	5.35***	3.95***	3.03***
	(0.52)	(0.37)	(0.51)	(0.44)	(0.36)
1 if Black	17.89***	14.54***	14.97***	15.98***	12.65***
	(1.18)	(0.97)	(1.16)	(1.12)	(0.98)
1 if Hispanic	9.95***	8.62***	6.37***	9.47***	6.52***
	(0.93)	(0.85)	(0.86)	(0.84)	(0.77)
1 if other race or ethnicity	3.12***	2.31**	3.90***	1.87*	2.26***
	(1.20)	(0.92)	(1.00)	(1.04)	(0.81)
1 if foreign-born	−1.53**	−1.43**	−0.12	−1.44**	−0.60
	(0.66)	(0.61)	(0.58)	(0.61)	(0.56)
1 if family income \$15,000 to \$30,000	–	−15.77***	–	–	−12.44***
		(1.11)			(0.96)
1 if family income \$30,000 to \$50,000	–	−24.09***	–	–	−18.47***
		(1.13)			(0.92)
1 if family income \$50,000 to \$75,000	–	−25.25***	–	–	−18.00***
		(1.13)			(0.88)
1 if family income \$75,000 or more	–	−25.65***	–	–	−17.18***
		(1.13)			(0.87)
1 if Some college	–	–	7.97***	–	4.09***
			(0.50)		(0.37)
1 if High school graduate	–	–	18.44***	–	12.82***
			(0.94)		(0.74)
1 if unemployed	–	–	–	18.92***	11.25***
				(1.46)	(1.08)
1 if not in the labor force	–	–	–	18.25***	8.79***
				(1.01)	(0.79)
Mean dep. var. (%)	10.28	10.28	10.28	10.28	10.28
Observations	24,955	24,955	24,955	24,955	24,955
Adjusted R^2	0.066	0.162	0.124	0.119	0.201

Note: I include income, education, and employment here only to show how sensitive the estimates are to controlling for them. Because these variables are measured only after high school, the mandate may already have affected them. Conditioning on them absorbs part of the treatment effect and attenuates the estimated gender differential. For this reason, I study them as outcomes in Table 7. All columns are ETWFE estimates with adoption-cohort and graduation-cohort fixed effects, estimated on the subsample with all three observed, and coefficients are in percentage points. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Event-study estimates of the effect of personal finance mandates on being unbanked by gender across estimators



Note: Estimates are shown separately for women and men. Each panel gives a different estimator: dynamic extended two-way fixed effects (ETWFE), Callaway–Sant'Anna, Sun–Abraham, and dynamic two-way fixed effects (TWFE). The first three use never-treated states as the comparison group; dynamic TWFE uses the standard TWFE comparisons. The horizontal axis is years since the state's first mandate-exposed cohort. Points are estimates in percentage points, and bars are 95 percent confidence intervals from state-clustered standard errors.